

Lab Project
EE 457 Fall 2024
Lab #5

Justin Pulley

For: Prof. Thuc

Date: 12/6/24

Purpose of Description

The purpose of this lab is to gain a clearer understanding of sampling theory and uniform quantization. And learn how to simulate these operations using MATLAB and Simulink

Design and Implementation Procedure

First, I used MATLAB to illustrate the sampling and reconstruction of a lowpass signal. I first created a message signal and sampled it by downsampling and upsampling. I calculated the fourier transform of the message signal and sampled the signal to see them in the frequency domain. And I created a low pass filter in the frequency domain.

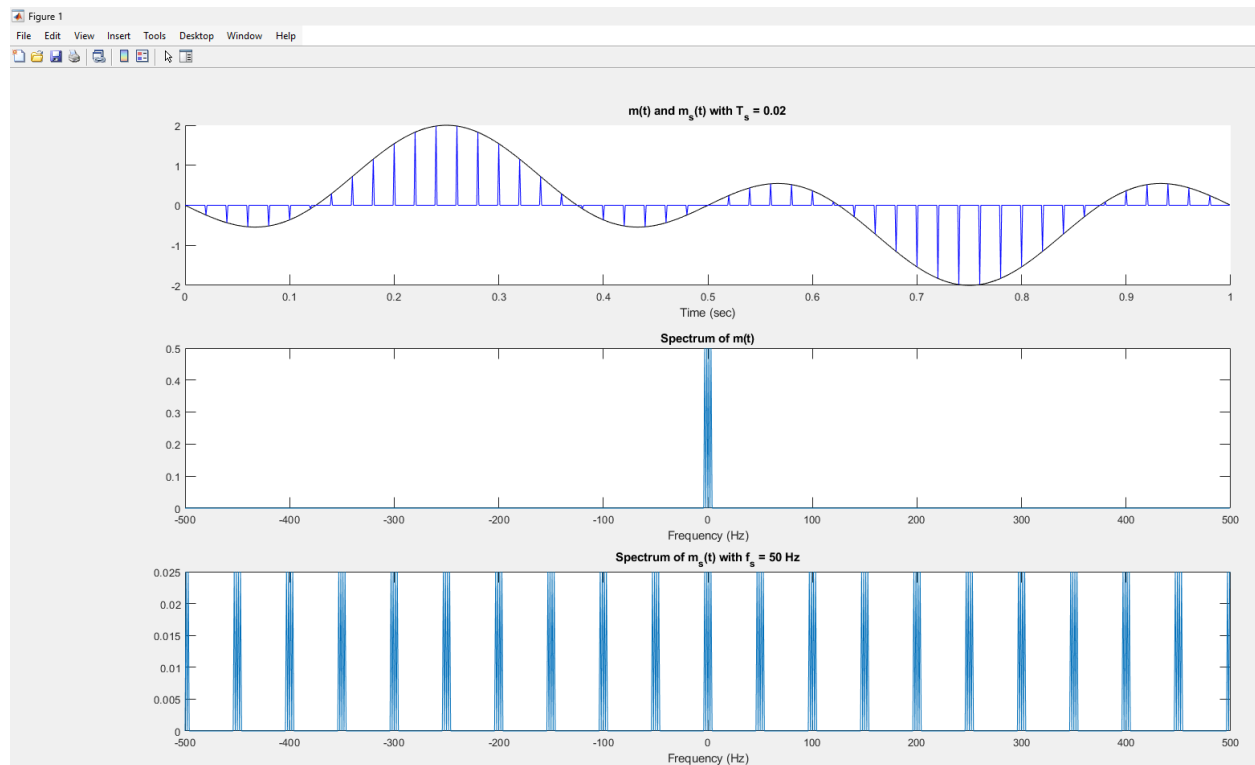


Figure 1: Message signal and sampled signal in time and frequency domain

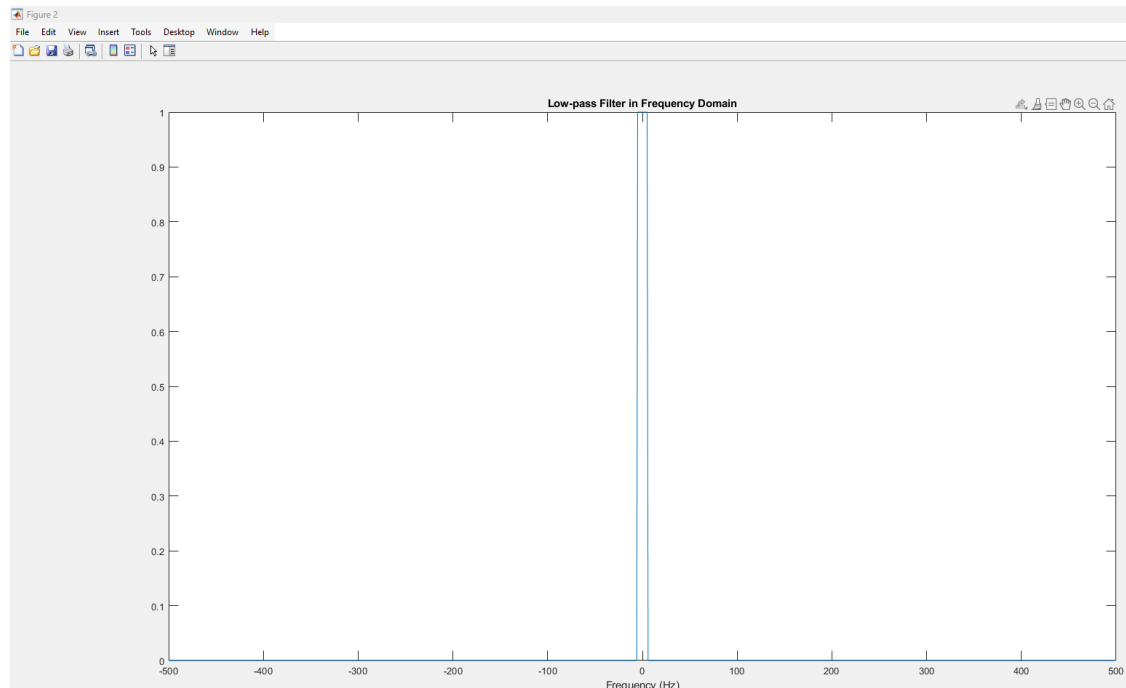


Figure 2: Low pass filter in the frequency domain.

Next, I ideally reconstructed the original signal by applying a low pass filter to the sampled signal with an equalizer. And getting the inverse fourier transform of the resulting signal.

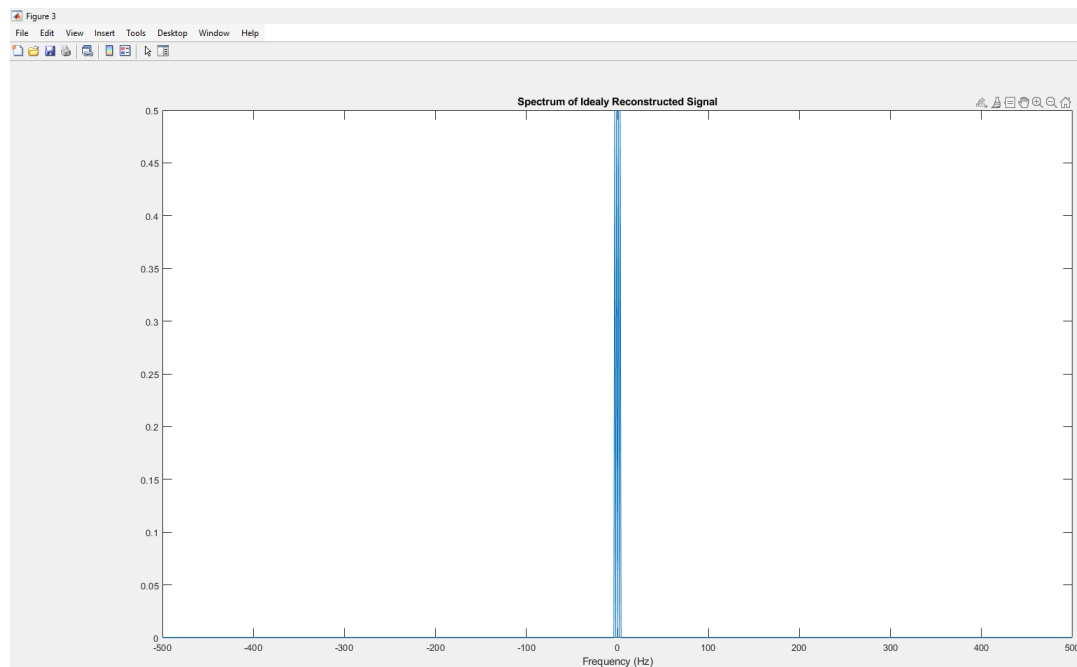


Figure 3: The ideally reconstructed signal in the frequency domain.

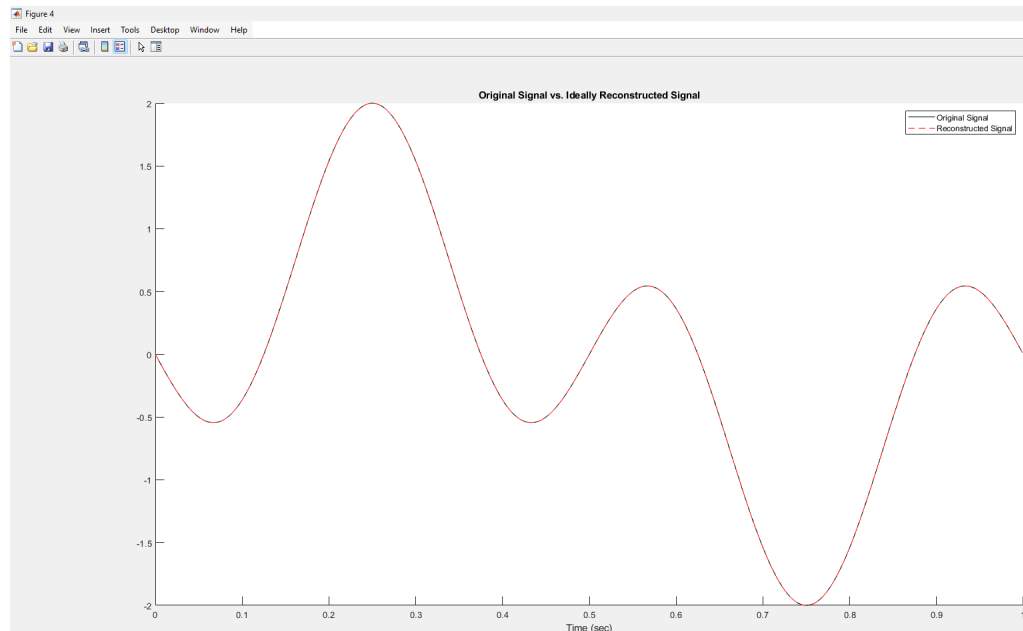


Figure 4: The original message signal vs. the ideally reconstructed signal in the time domain.

Next, I used flat-top sampling to sample the original message signal. Then I took the fourier transform of the sampled signal. I applied a low-pass filter and took the inverse fourier transform to get another reconstructed signal.

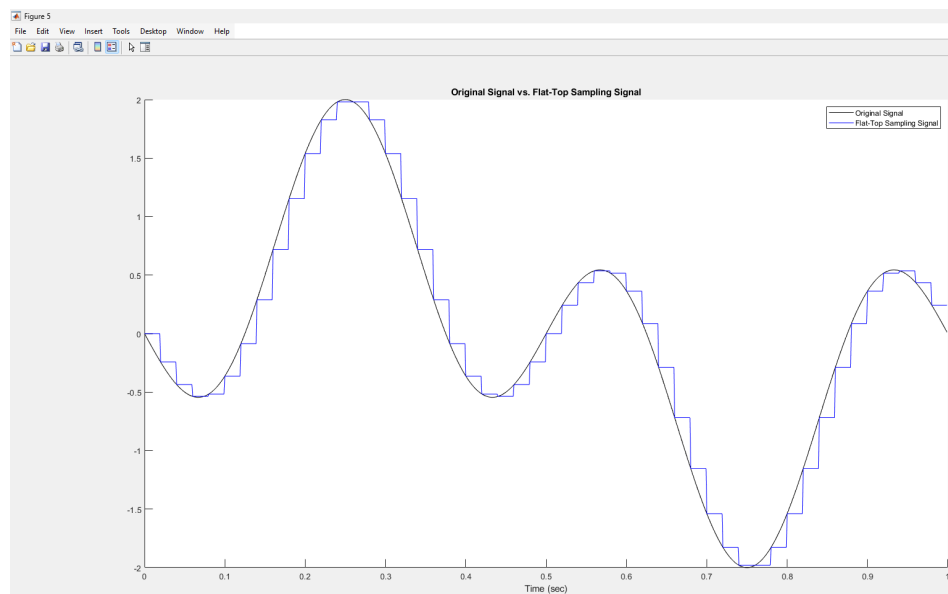


Figure 5: The original message signal and flat-top sampled signal in the time domain

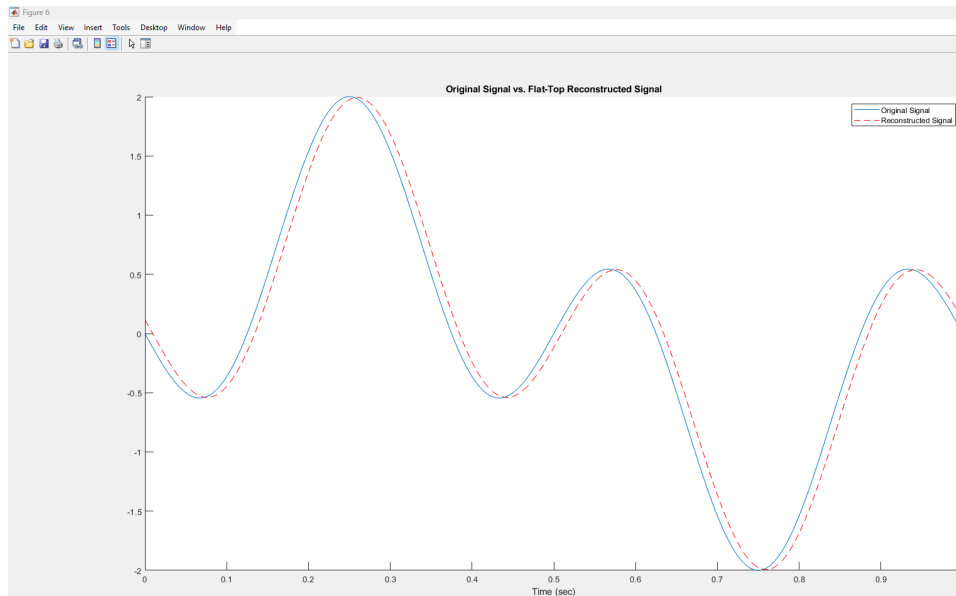


Figure 6: The original message signal vs. the flat-top reconstructed signal in the time domain.

After that I downsampled the message signal. Then I quantized the downsampled message signal at varying set quantization levels: 4,8,16,32. I calculated the signal to quantization noise ratio (SQNR) associated with each set of quantization levels. As the quantization levels increased the SQNR increased.

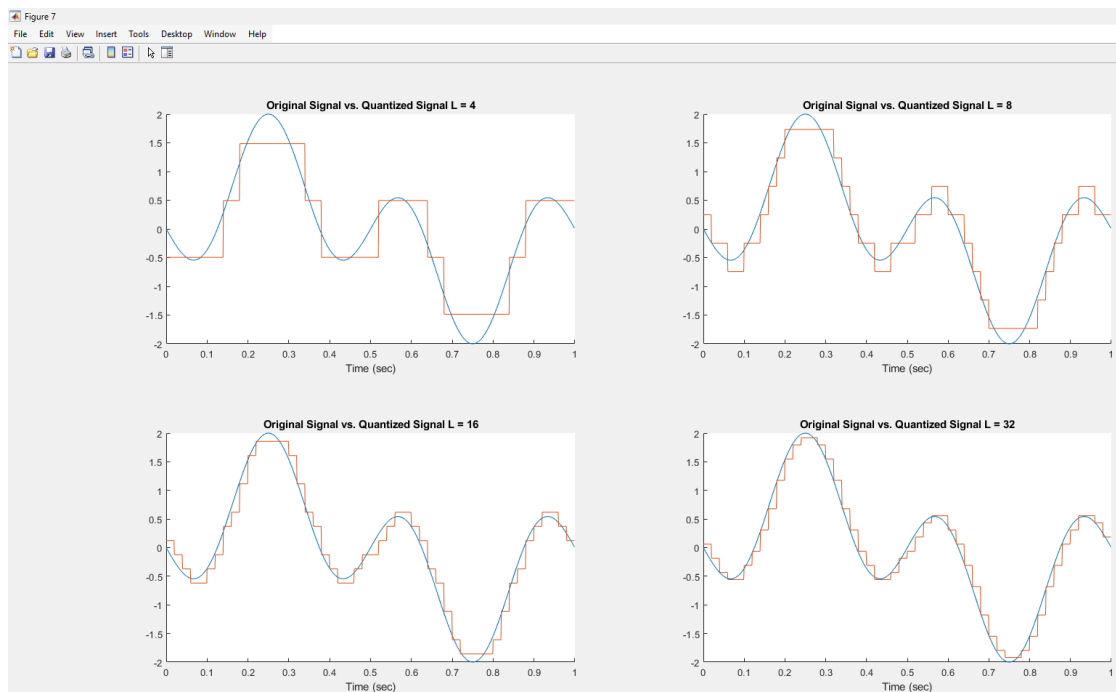


Figure 7: The original message signal vs. the quantized signal at varying quantization levels.

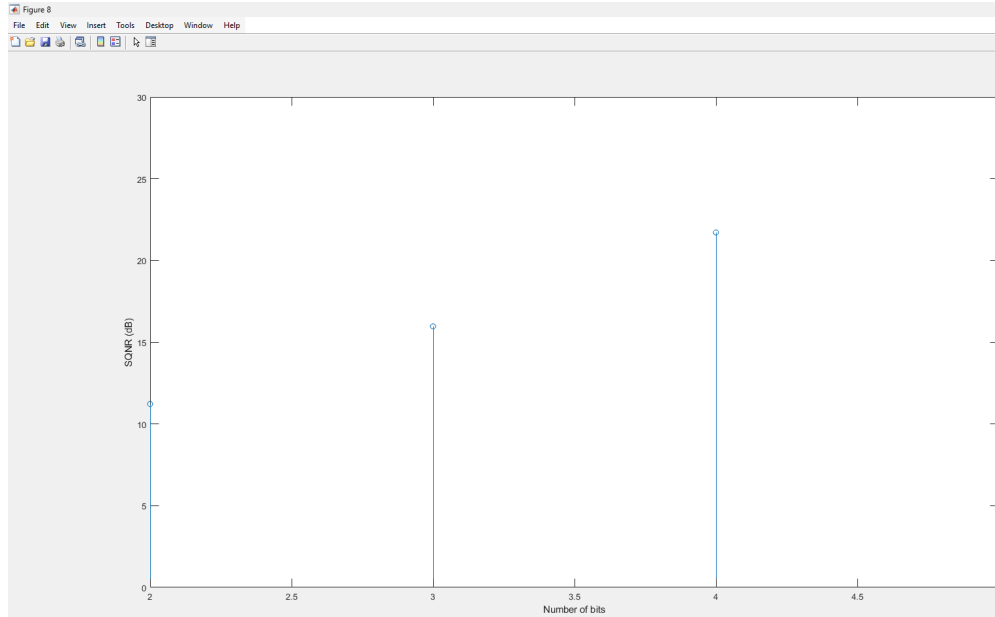


Figure 8: Number of bits vs. the signal to quantization noise ratio.

Finally, I built a simulink model of an analog to digital converter (ADC) system.

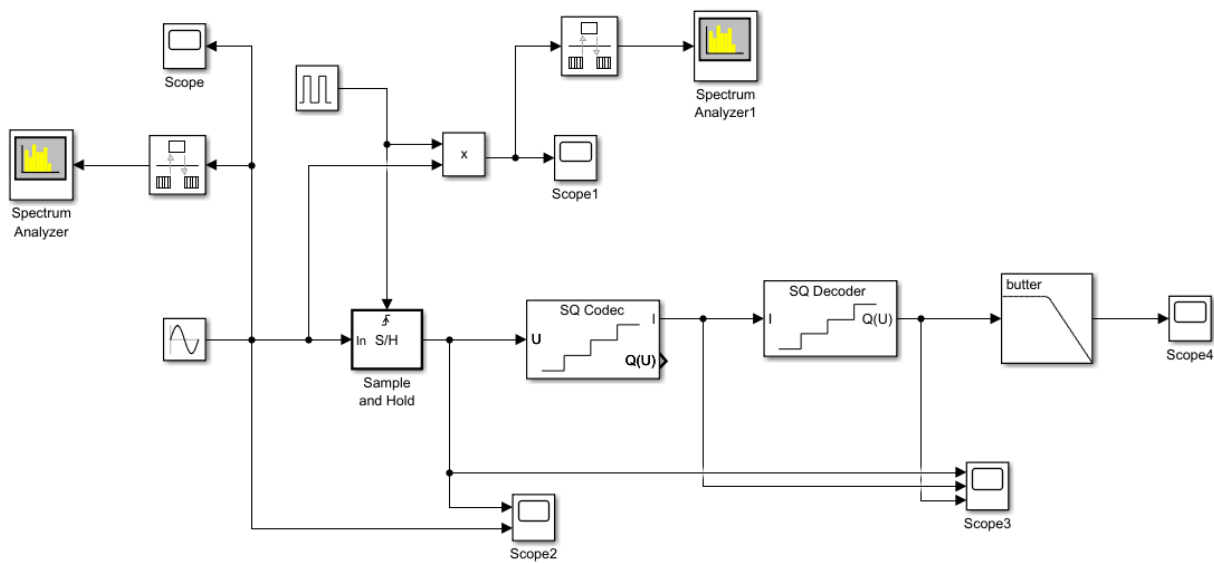


Figure 9: Simulink model of an analog to digital converter.

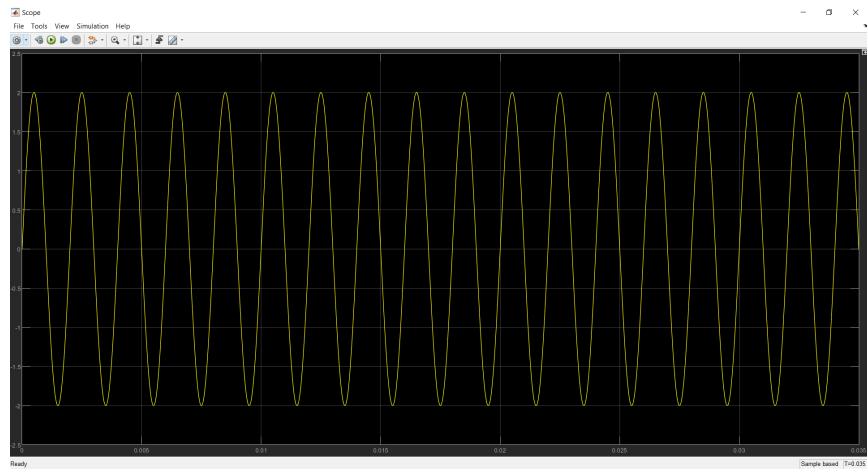


Figure 9.1: Output of Scope from the simulink model of an analog to digital converter.

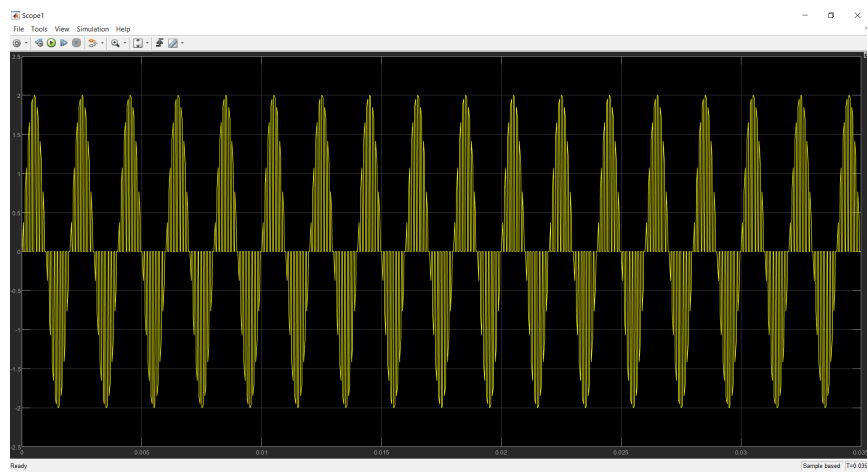


Figure 9.2: Output of Scope2 from the simulink model of an analog to digital converter.

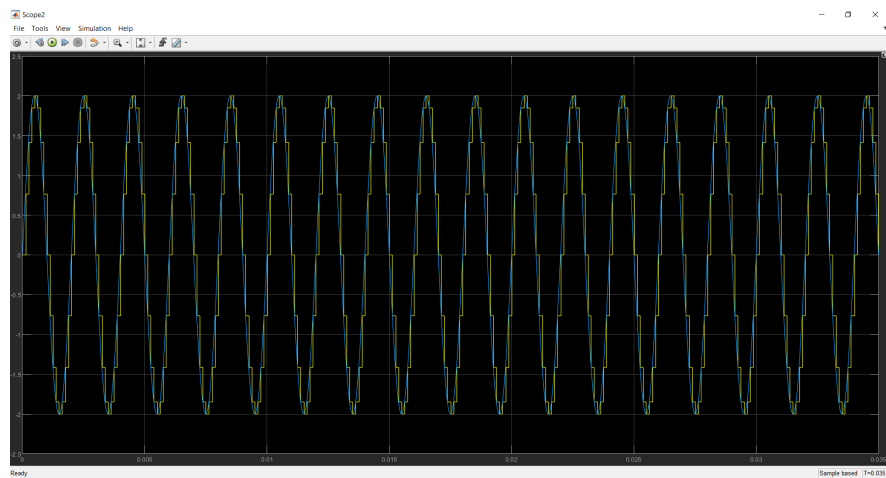


Figure 9.3: Output of Scope2 from the simulink model of an analog to digital converter.

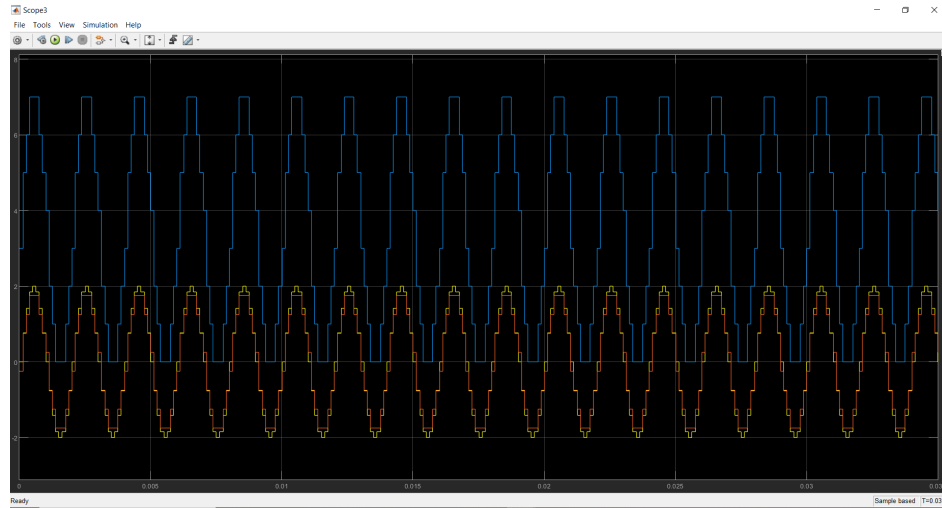


Figure 9.4: Output of Scope3 from the simulink model of an analog to digital converter.

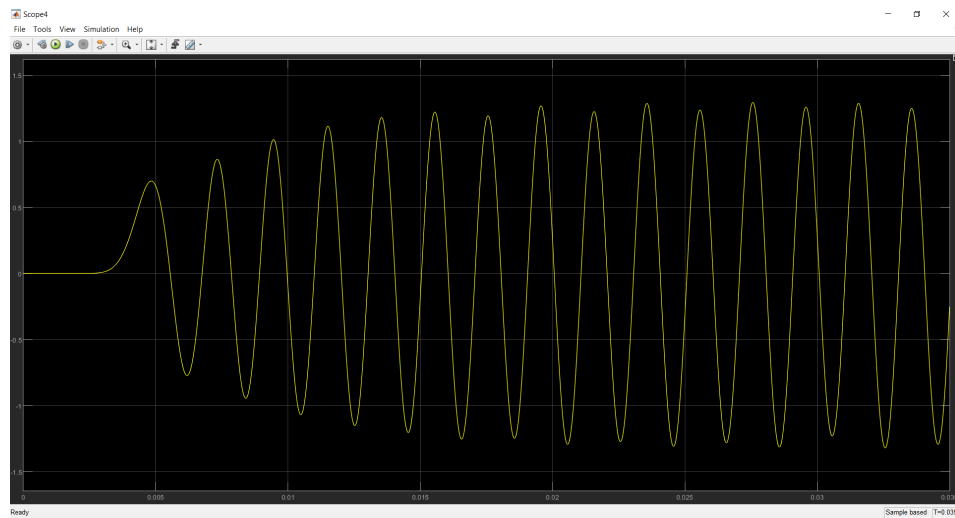


Figure 9.5: Output of Scope4 from the simulink model of an analog to digital converter.

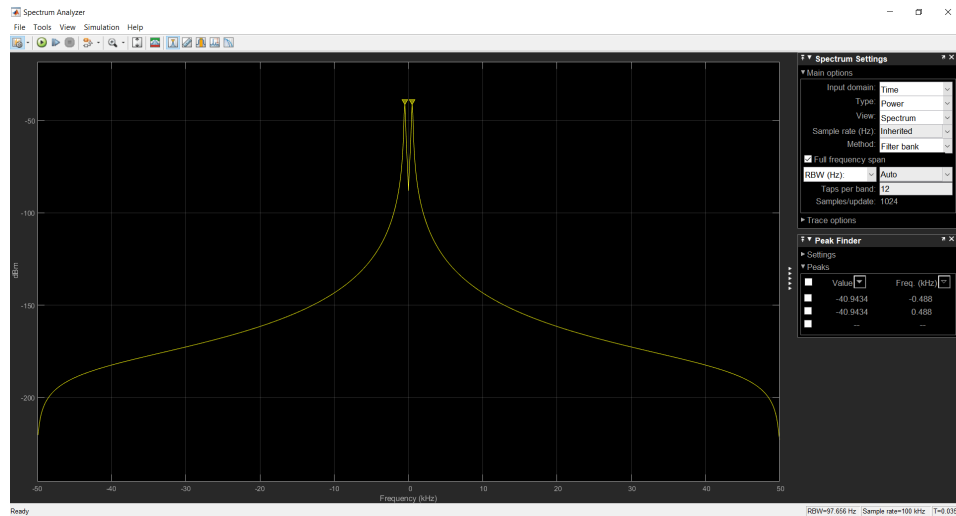


Figure 9.6: Output of Spectrum Analyzer from the simulink model of an analog to digital converter.

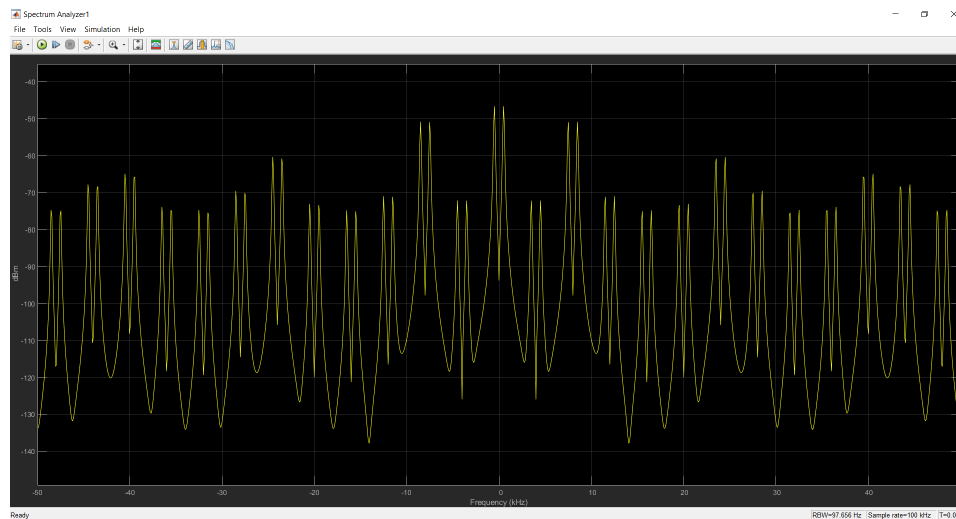


Figure 9.7: Output of Spectrum Analyzer1 from the simulink model of an analog to digital converter.

Lessons Learned

Throughout this lab, I learned the sampling theory and uniform quantization. And learn how to simulate these operations using MATLAB and Simulink. I learned how to ideally and flat top sample. How to reconstruct a message signal after it has been sampled. And I learned how to build an ADC in Simulink.